Speeding up Feynman integral reduction with Kira: exact identities and learned choices

Rushil Saraswat

October 20, 2026

Abstract

Feynman integral reduction expresses the many loop integrals in a particle-physics calculation through a smaller set of master integrals. We study how to shorten this step using Kira, exact analytic identities, and machine learning. The method first checks whether an integral has a direct formula or a recurrence that reduces its propagator powers. It sends the remaining targets to Kira, where measured reduction costs determine which integrals generate equations and how the coefficients are calculated. We test each added learned choice against a fixed procedure and retain it only when validation supports its use. The final comparison contains 93 requests, including connected two- and three-loop families and restricted analytic families through thirteen loops. Median time for the complete collection falls from 45.091 to 41.137 seconds relative to the preceding application, a reduction of 8.8%. All 837 results in the three-method, three-repetition comparison pass the exact reference comparisons. Earlier seed and equation models give gains on their own test sets, but the added arithmetic selector and equation filter do not improve the final method and remain disabled. The results establish a faster combined procedure for the stated inputs. They leave general connected high-loop scattering and a reliable additional machine-learning gain in the final comparison unverified.

1 Introduction

Predictions for particle scattering and decay involve amplitudes, whose squared magnitudes enter measurable probabilities and rates. In perturbation theory, an amplitude is calculated as an expansion in the interaction strength. Higher-order terms contain integrations over internal momenta that are not fixed by the external particles. These integrations give Feynman integrals, and the number of independent internal momenta is the number of loops [1]. Quantum chromodynamics, the theory of the strong interaction, requires these integrals in higher-order perturbative corrections to scattering amplitudes and particle decay amplitudes [1].

Many integrals share the same denominator factors and differ only in their powers or numerators. Integration-by-parts (IBP) identities relate such integrals, so the calculation can express them through master integrals: a set of integrals that remains after the chosen reduction relations have been used [2, 3]. Kira automates the construction and solution of these relations [4]. Finding the coefficients of the masters is the reduction problem studied here. Evaluating the remaining masters is a further step, although some of the analytic formulas used below also provide their values.

The question is how to complete the requested reduction with less elapsed time. Kira already has methods for restricting equation generation and removing unnecessary relations [5]. A further restriction can save work, but an insufficient equation system requires recovery. Our study therefore compares complete calculations, including the choice of method and any repeated attempt. Machine-assisted seed search provides a related approach to reducing the equation system [6]; analytic reduction provides another route when the denominator factors satisfy suitable conditions [7, 8].

We combine these routes and measure their contributions separately. Section 2 defines the integrals and explains how Kira reduces them. Section 3 describes the exact formulas and learned choices, followed by the comparison design in Section 4. Sections 5 and 6 state the measured gains, slower cases, and range of application. The appendices give the derivations and physical inputs; the companion notes retain the full sequence of experiments.

2 From a loop integral to a master-integral reduction

We first define the notation used in this paper. This makes it possible to state both the reduction problem and the conditions under which an exact shortcut applies.

2.1 What the integrals contain

We use the normalized integral

$$I_{\boldsymbol{a}}(\boldsymbol{x}, d) = \int \prod_{h=1}^{L} \frac{d^d \ell_h}{i\pi^{d/2}} \prod_{j=1}^{N} D_j^{-a_j}, \qquad D_j = Q_j^2 - m_j^2 + i0, \qquad d = 4 - 2\epsilon. \tag{1}$$

Here L is the number of loops, ℓ_h is an integration momentum, and N is the number of denominator coordinates. Each Q_j is a specified linear combination of loop momenta and external momenta, and m_j is a mass. The vector $\boldsymbol{x}$ contains squared masses and scalar products of external momenta. The symbol i is the imaginary unit, and $i0$ means an infinitesimal positive imaginary part that specifies the Feynman pole prescription. An external invariant is a scalar product that does not change between inertial reference frames. For a physical internal line, $1/D_j$ is its scalar propagator and D_j is its inverse. The normalization $i\pi^{d/2}$ is conventional. Analytic continuation extends an integral formula from a region where the integral converges to other parameter values. Dimensional regularization uses this continuation in d; its poles describe divergences as the regulator ϵ approaches zero [1].

The integer vector $\boldsymbol{a} = (a_1, \ldots, a_N)$ identifies an integral. A positive a_j places $D_j^{a_j}$ in the denominator, $a_j = 0$ removes that factor, and a negative a_j places a power of D_j in the numerator. An *integral family* fixes all Q_j, masses, and external scalar products while allowing these indices to vary. Auxiliary coordinates represent numerator scalar products that the physical denominators do not span [4]. Thus a family specifies the mathematical problem before any model makes a choice.

For example, a one-loop two-point integral, commonly called a bubble, has the form

$$B_{ab}(p^2; x, y) = \int \frac{d^d \ell}{i\pi^{d/2}} \frac{1}{(\ell^2 - x + i0)^a [(\ell - p)^2 - y + i0]^b}. \tag{2}$$

This equation defines x and y as squared masses, p as the external momentum, and a, b as denominator powers. The integrals B_{11} and B_{21} belong to the same family and have different powers of the first denominator. More complicated families introduce additional loop momenta and denominator factors, using the same notation.

The chosen set of masters is called a master basis. A *target* is an integral whose reduction is requested. A *sector* groups integrals with the same positive-index positions, and a sub-sector removes one or more of those positions [4]. A request in our experiments contains one family, its physical parameters, and one or more targets. This distinction matters because a single timing measurement can cover several targets.

2.2 How integration by parts relates the targets

Integration by parts supplies an equation by integrating a total derivative. A seed is an integer index vector at which we generate these identities. For seed indices $\boldsymbol{\sigma}$ and a loop or external momentum v, the identity is [2]

$$0 = \int \prod_h \frac{d^d \ell_h}{i\pi^{d/2}} \frac{\partial}{\partial \ell_k^\mu} \left(v^\mu \prod_j D_j^{-\sigma_j} \right) = \sum_u A_{\boldsymbol{\sigma},o;u}(\boldsymbol{x}, d) I_{\boldsymbol{a}^{(u)}}. \tag{3}$$

The repeated component index μ is summed, k selects the differentiated loop momentum, and $o = (k, v)$ labels the derivative operation. Expanding the derivative changes denominator powers, giving the coefficients $A_{\boldsymbol{\sigma},o;u}$, where u labels the resulting integrals. Appendix A derives these index shifts. Analytic continuation gives the dimensionally regularized identity outside an initial convergence region [2].

A simple example shows how an identity becomes a reduction. A vacuum integral has no external momentum entering its denominators. Define its one-denominator example, called a tadpole, by

$$T_a(x) = \int \frac{d^d \ell}{i\pi^{d/2}} (\ell^2 - x + i0)^{-a}, \qquad x \neq 0. \tag{4}$$

Applying Eq. (3) with $v = \ell$ gives

$$0 = (d - 2a) T_a(x) - 2ax\, T_{a+1}(x), \qquad T_2(x) = \frac{d-2}{2x}\, T_1(x). \tag{5}$$

Appendix A shows the intermediate derivative. The last relation expresses the target T_2 through the master T_1. Its coefficient depends on d and the squared mass; reduction coefficients need not be constants.

2.3 What Kira calculates and where the work occurs

A seed is the index vector $\boldsymbol{\sigma}$ used in Eq. (3), and each seed supplies several derivative equations. Kira constructs a finite system, identifies relations using symmetry and linear dependence, and eliminates integrals according to an ordering. Back-substitution then writes the requested targets in terms of the remaining masters [4, 3]. The desired result is

$$I_{\boldsymbol{a}} = \sum_{m=1}^{n_M} r_{\boldsymbol{a}m}(\boldsymbol{x}, d) M_m, \qquad r_{\boldsymbol{a}m} \in \mathbb{Q}(\boldsymbol{x}, d). \tag{6}$$

Here M_m is master m, n_M is the size of the chosen master set, and $\mathbb{Q}(\boldsymbol{x}, d)$ denotes rational functions with rational coefficients. Solving the linear identities gives this form, as shown in Appendix A. A smaller seed system can leave extra free integrals, so its remaining columns alone do not establish a complete master basis.

There are three distinct choices that a learned model can address: which seeds generate equations, which equations to retain, and which allowed row operation to perform next. A row is one equation in the coefficient matrix; an elimination step uses a nonzero coefficient to remove an integral from other rows. Appendix A derives that operation. Table 2 gives the models studied at these three locations and explains which remain active.

The coefficient arithmetic creates another cost. Symbolic elimination manipulates rational functions throughout the calculation. Alternatively, Kira can evaluate the system at numerical points modulo a prime and use FireFly to reconstruct the rational functions [9, 10, 11]. Such a finite field uses exact arithmetic on residues, so the numerical samples are not floating-point approximations. Appendix L explains how samples constrain a coefficient. Our method can choose this existing arithmetic procedure, in addition to changing the equations or applying a direct identity.

3 A combined method for reducing the total time

Our procedure first tests the mathematical conditions for an exact identity. It applies an accepted formula to a whole target or to a reducible part, then uses Kira for the remaining integrals. Within that calculation, a learned choice is used only in the domain where its validation supports it. This ordering gives each method a defined role and allows a direct comparison with the same calculation performed without the additional learned choice.

3.1 Exact formulas and recurrences

The tadpole relation in Eq. (5) illustrates the first route: a recurrence repeatedly lowers an integer power. A direct formula can go further and provide an analytic value. In our normalization, Appendix I derives

$$T_a(x) = (-1)^a (x - i0)^{d/2-a} \frac{\Gamma(a - d/2)}{\Gamma(a)}. \tag{7}$$

The Gamma function is defined for $\operatorname{Re} z > 0$ by $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt$ and elsewhere by analytic continuation [12]. It satisfies $\Gamma(z+1) = z\Gamma(z)$, so integer shifts of its argument give finite products. These products explain why the ratios of the target and master values in our analytic families are rational.

Some formulas depend on a particular physical limit. Feynman parametrization combines denominator factors through an auxiliary parameter integral. For the bubble of Eq. (2) at $p^2 = 0$, this procedure gives

$$B_{11}(0; x, y) = \frac{T_1(x) - T_1(y)}{x - y}, \qquad B_{ab}(0; x, y) = \frac{\partial_x^{a-1} \partial_y^{b-1} B_{11}(0; x, y)}{(a-1)!(b-1)!}. \tag{8}$$

The first expression assumes $x \neq y$. At equal nonzero masses the limiting result is $B_{ab}(0; x, x) = T_{a+b}(x)$. Appendix G derives both cases and the extension to polynomial numerators. This separate formula avoids dividing by a matrix of external scalar products that becomes singular at this kinematic point.

The latest exact addition concerns a massless triangle with three denominator powers a, b, c, two lightlike external momenta p_1, p_2, and one nonzero invariant $Q = -(p_1 + p_2)^2 > 0$. Lightlike means $p_1^2 = p_2^2 = 0$. Appendix M defines the integral $T_{abc}(Q, d)$, derives its Gamma-function representation, and converts it to a bubble master. In particular,

$$T_{111}(Q, d) = \frac{2(d-3)}{Q(d-4)}\, B(Q, d), \qquad B(Q, d) = T_{101}(Q, d). \tag{9}$$

This is a standard analytically tractable kinematic configuration [13]. We derive the coefficient with our sign and normalization conventions before using it. The preceding application already reduced this triangle analytically to a triangle master; the addition supplies the bubble-basis coefficient and an explicit master value.

The same reasoning extends to selected many-loop structures. If separate groups of loop momenta occur in separate factors of the integrand, their reductions multiply. Some connected families also allow successive bubble integrations from the outermost loop inward. Appendix H defines these trees and derives their Gamma products. Appendix A proves the independent-product rule. The accepted domains are summarized in Table 1; their restrictions prevent the formulas from being applied to a different physical problem.

For clarity, a scalar denominator integral has no free Lorentz index. Polynomial numerators arise after contractions of tensor integrals, whose numerators carry momentum components. We use angular averages to reduce these numerator terms; Appendix F derives the required contractions. An integral is scaleless when it has no nonzero mass or external invariant that can supply

Table 1: Exact methods used in the application. These are implementation domains, with derivations in the stated appendices. "Rank" is the total numerator power, and "excess power" counts positive powers beyond one.

Integral structure	Method	Principal conditions
One-loop scalar and polynomial-numerator sectors	Angular reduction and a power-lowering recurrence (Appendix F)	At most five coordinates; rank and excess power at most six; required matrices invertible
Two-point sectors at $p^2 = 0$	Endpoint tadpoles and mass derivatives (Appendix G)	Rational tensor inputs: rank and excess power at most four; scalar symbolic masses: individual powers at most six and total at most eight
Vacuum factors with one momentum per factor	Partial fractions and tadpole values (Appendix I)	Nonnegative rational squared masses; at most eight denominators per factor; powers bounded as stated there
Scaleless vacuum families	Zero in dimensional regularization (Appendix I)	No mass or external momentum scale; homogeneous quadratic denominators
Connected bubble trees	Successive exact bubble integrations (Appendix H)	Massless two-point root or one massive vacuum root; no additional occupied auxiliary lines
Massless triangles	Direct coefficient and Gamma value (Appendix M)	Two lightlike legs; rational $Q > 0$; powers 0–4 and total at most eight
Independent products	Product of component reductions (Appendix A)	Every occupied factor belongs to an accepted independent component

a dimensionful scale. Appendix I derives its zero value in dimensional regularization. The implementation admits the listed vacuum, tree, and independent-product structures through thirteen loops. This loop range describes specific structures. Additional propagators that couple their loop momenta can invalidate the factorization or the tree integration. Unsupported inputs therefore continue to the remaining exact rules or Kira, subject to the application's resource bounds.

3.2 Learned choices within Kira

Machine learning enters as a prediction of a useful computational choice. Training fits a model to examples, validation selects its settings on separate examples, and testing evaluates the frozen choice on reserved inputs. A model receives quantities calculated from the target indices and propagators, called features, and estimates the cost or usefulness of an allowed action. The algebra still determines the reduction coefficients. A decision tree splits its inputs through a sequence of feature tests and returns a prediction at the final branch [14]. Table 2 separates the original three interventions. Here and below, *native* means that Kira generates and solves the IBP system.

Table 2: The three learned interventions and their current use. Their separate timing tests appear in Table 6; the notes give the earlier architectures and training histories.

Choice	Model and input	Predicted action	Final application
Seed selection	A depth-two cost tree, with twelve depth-three failure trees per action; family and target features	One of three seed regions	Retained in its accepted domain
Equation selection	A decision-tree classifier; seed, target, and derivative features	A retention score for each seed–operator pair	Disabled in the final comparison
Row selection	A set-based neural ranker for the current matrix and allowed steps; later a compiled cost policy	The next elimination step	Omitted; native row rules retained

The retained seed model selects among wide, medium, and tight regions of seed indices. It was trained on 34 batches and selected using sixteen separate validation batches. Appendix B defines the regions. They are candidate searches, so a predicted choice does not prove that its equations suffice.

The equation model has seventeen features that describe a candidate seed, its derivative operator, and its relation to the targets. Its training labels identify equations retained by a complete reference reduction. A score threshold selects equations before their full coefficients are constructed. Appendix B specifies the features and weighted training labels. The original row model uses a neural network: a fitted sequence of numerical maps that assigns each candidate step a score. It combines 132 candidate features through shared 32-component representations, their mean and maximum, and a map with 64 intermediate components. Combining a set of candidates in this way follows the set-based representation of Ref. [15]. The notes give its training on constructed target-row reductions and its subsequent test inside Kira's changing matrix. A row model acts when the matrix already exists, and must save enough algebra to cover the cost of choosing each step. The measured row results did not support its use in the final application.

3.3 Learning the cost of the coefficient arithmetic

We also tested a model that chooses the arithmetic for the remaining native reductions. The candidate actions are the preceding procedure and finite-field reconstruction with 1, 4, 8, or 16

evaluation points processed together. This group size is separate from the number of target integrals in a request.

The latest training collection contains 48 requests across four propagator structures, with 20 further requests for validation. A three-loop training comparison did not agree with its reference; later checks did not reproduce the discrepancy. We therefore keep the preceding procedure for that structure and fit the added model to the 36 two-loop training requests. The three-loop requests remain in the complete comparison. The notes preserve the initial failure and the subsequent checks.

The fitted model averages 32 extremely randomized regression trees, using the construction of Geurts, Ernst, and Wehenkel [16]. Its 29 features describe the loop count, masses, denominator connections, target powers, and sizes of the supplied rational kinematic numbers. It predicts complete time relative to the same request's preceding procedure. Appendix C derives the weighted training objective and lists the features.

Validation selects tree depth from 2, 4, and 6, minimum terminal-branch population from 2 and 4, and the required predicted saving from zero, five, and ten percent. A fixed control chooses its best setting separately for each of the four structures. The model is enabled only if it reduces validation cost by at least three percent and selects no failed procedure. This is an experimental acceptance condition, not a mathematical guarantee for unseen integrals. The model fails this condition in the recorded comparison, so the final application uses the fixed settings and keeps the earlier seed model.

3.4 Completion, recovery, and shared work

An accepted exact formula requires its stated propagator and kinematic conditions. A restricted native calculation must express every target in the required master basis. When it does not, the application attempts its configured broader or preceding procedure and includes the spent time in the result. In the experiments, a separate exact comparison also subtracts every returned coefficient from the reference coefficient in the same basis. This additional check tests the actual output; a model score alone supplies no correctness certificate.

The application also processes a list of requests in one call. This shares initialization and repeated exact component calculations across the list. It preserves the factorized form of independent products instead of expanding all products of master terms. The final timings include this shared-call organization in both the new method and its control, as explained next.

4 Physical scope and comparison design

We use complete reduction requests to test the method, since fewer seeds or a more accurate label prediction does not determine the elapsed time. This section states what the requests contain, which calculations serve as controls, and what enters each timer. All numerical results below are measurements from this study; external references support the established methods, while the appendices derive the identities and estimators used here.

4.1 The final collection

The final collection has 93 requests. It combines 40 new requests with 53 previously checked analytic requests that test whether the existing exact methods still work. Table 3 identifies the physical structures. The exact propagators, master lists, target vectors, and kinematic assignments are given in Appendices D, J, K, and N.

The names planar and nonplanar refer to the line-connection structures of the two box families. Their equations in Appendix D specify the exact meaning used here. Both structures appear in

Table 3: The 93 requests in the final comparison. A request can contain several targets. Native inputs use rational masses and external invariants with symbolic dimension d.

Structure	Loops	Requests	Reduction and scope
Planar and nonplanar boxes	2	$5 + 5$	Four external legs; native IBP reduction
Massive sunrise	2	5	Two-point family with three massive physical lines; native IBP reduction
Massive vacuum family	3	5	Six coupled physical denominators; native IBP reduction
Massless triangles and independent products	1–13	20	One nonzero invariant per triangle; direct formula
Independent vacuum products	1–13	13	Two squared masses per component; partial fractions
Connected one-mass vacuum trees	1–13	34	Chain, star, and binary-tree connections; successive integrations
Zero-invariant and ordinary one-loop numerator integrals	1	$3 + 3$	Tensor recurrences and the endpoint formula
Total		93	20 native and 73 analytic requests

training, validation, and testing. The new native tests change the physical points and target batches, so they measure transfer within known structures. They do not hold out every graph structure. Individual index patterns can also recur between splits.

For the new box inputs, training uses indices $i = 2, \ldots, 13$, validation uses $i = 14, \ldots, 18$, and testing uses $i = 19, \ldots, 23$, with the physical assignments in Appendix N. Targets include numerator powers and raised physical denominator powers. The independent triangle and vacuum products include every loop order from one through thirteen. Connected tree tests cover the same loop range under their specified one-scale assumptions.

4.2 Controls and time measurements

The final experiment compares three applications on exactly the same 93 requests. The *preceding application* is the published method before the direct triangle addition and its new native-setting selection. The *intermediate application* adds those changes with the original runtime monitor. The *final application* also corrects the monitor's handling of temporary entries that disappear while a reduction runs. The inherited seed model is shared by all three. The added arithmetic selector and equation filter are disabled by validation in the final method.

The native implementation uses Kira 3.1 with the retained project reduction choices; its reconstruction-enabled build includes FireFly. Each application processes the collection in one call. We use one worker on the same Intel Gold compute node, one computational thread per application, and three repetitions. We rotate method order and reverse request order in the middle repetition. The measured time includes loading, method selection, algebra, reading outputs, cleanup, and actual recovery. Preparation of independent reference reductions and the external exact coefficient comparisons lie outside every application timer. The comparison therefore measures the reduction application, with an additional untimed correctness audit.

We report the median of the three complete-collection times and their observed minimum and maximum. These three-observation ranges are not confidence intervals. For diagnostic results within a physical group, we sum the per-request median times recorded inside the shared call. These component sums omit some shared overhead and are not the complete-collection estimator. Appendix E states both definitions.

The native references are independently prepared symbolic reductions in the prescribed bases. Independent symbolic partial fractions check the vacuum products. Tree checks compare the

retained recurrence with the preceding exact result and its analytic derivation. Triangle checks use exact conversion between the preceding triangle basis and the new bubble basis, separate one-loop Kira references, and Gamma-function identities. Analytic products are justified by the factorization proof. The notes state the individual check counts and preserve the unsuccessful trials.

4.3 Separate comparisons needed to identify the cause of a gain

The final baseline already contains exact formulas, learned seeds, and earlier Kira improvements. Its time ratio therefore measures the latest combined change; it does not measure a speedup relative to unmodified Kira. We retain three other comparisons to identify what the individual methods contribute.

First, a fresh-call comparison measures 40 new requests separately, including a new process for each request. Second, an earlier 75-request comparison isolates the addition of direct vacuum formulas and validated native reconstruction settings on one matched workload. Third, model-on/model-off comparisons hold the other procedures fixed. An *ablation* is such a comparison with one component removed. We use it to distinguish learned savings from analytic formulas, fixed tuning, and shared work.

The older fresh-call comparisons use sums of per-request median times. Where stated, their uncertainty intervals resample paired requests 5,000 times and report the central 95% of the resulting ratios, following the bootstrap method [17]. This measures variation across the sampled requests. It gives no confidence statement for all possible integral families. We report each workload and control beside its result and do not multiply ratios from different tests.

5 Results

The combined method reduces the time for the final collection, but the contribution depends on the physical structure and on how the requests are processed. We first report the complete comparison, then the individual families and learned components.

5.1 Complete time for the final 93 requests

The final application takes a median of 41.137 seconds, compared with 45.091 seconds for the preceding application. The ratio is 0.912, which corresponds to 8.8% less time. Table 4 gives all three measured procedures, and Figure 1 shows each repetition. All 837 application results, from 93 requests, three methods, and three repetitions, pass the 13,401 exact coefficient comparisons. No internal native attempt fails in this final serial comparison.

Table 4: Complete-collection elapsed time for 93 matched requests. The ranges contain the three observed times; they are not confidence intervals. All methods process the collection in one call.

Application	Median (s)	Observed range (s)
Preceding	45.091	44.783–45.817
Intermediate	42.932	40.351–44.589
Final	41.137	40.723–41.227

The intermediate-to-final difference is smaller than the preceding-to-final difference and varies with the execution conditions. The monitor correction removes a reproduced interruption mechanism, but these timings alone do not assign all of their difference to that correction. The scientific result is the measured total change of the combined procedure.

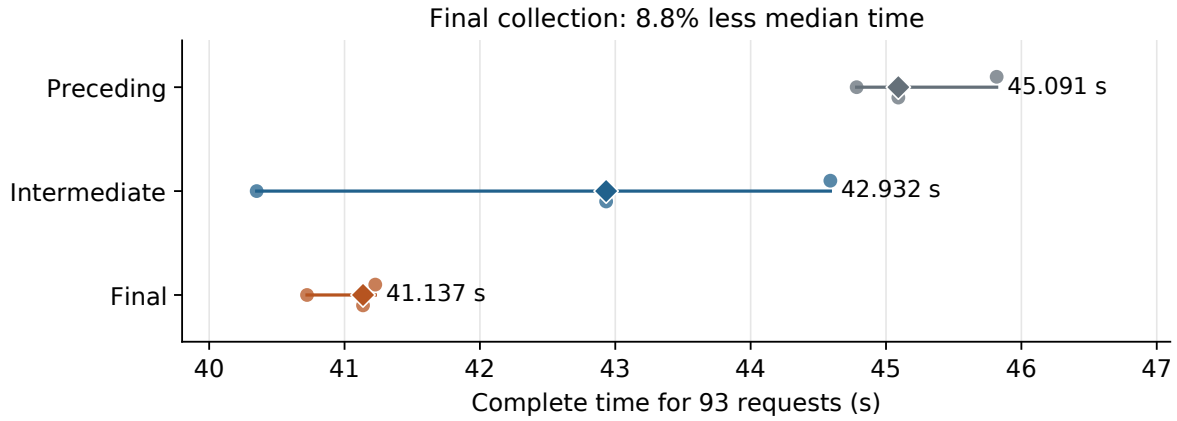


Figure 1: The final method reduces complete-collection time in each recorded repetition. Points are the three observations for each method; diamonds are their medians. The elapsed-time axis is restricted to the observed range so that the comparison remains readable. All points include shared setup and recovery costs.

5.2 Which integral structures become faster

Table 5 shows the native groups within the final shared-call experiment. Planar and nonplanar boxes become faster, while the sunrise group becomes slower and the three-loop vacuum group changes little. Thus an aggregate improvement does not imply an improvement for every family. The component times also show that the remaining native algebra dominates this particular collection.

Table 5: Native component times inside the final shared-call comparison. Each entry is a sum of per-request medians for five requests. Ratios divide final time by preceding time; shared overhead is excluded from these component sums.

Structure	Preceding (s)	Final (s)	Ratio
Planar box	8.948	7.191	0.804
Nonplanar box	14.517	11.137	0.767
Massive sunrise	7.509	8.407	1.120
Three-loop vacuum	12.971	12.988	1.001

The separate fresh-call experiment asks a different question: how much time each new request needs when it starts its own application. Across its 40 requests, summed median time falls from 57.766 to 50.848 seconds, a ratio of 0.880. The paired request-resampling interval is $[0.744, 0.992]$. The 20 native requests alone give 0.887 with interval $[0.743, 1.010]$, which includes equal time. These results refer to the direct triangle and native-setting changes before the final monitor correction.

Figure 2 shows the fresh-call change by physical structure and the triangle-product points at each loop order. The triangle group gives a ratio of 0.816, but the seven-loop product gives 1.077 and is slower. Higher-loop products test the factorization identity under the specified powers and invariants. They supply no timing result for a general connected scattering graph at the same loop order.

5.3 How much of the gain comes from machine learning

Earlier controlled tests resolve small seed-selection and equation-selection gains on their stated workloads (Table 6). These results motivated the learned interventions. The row model took more

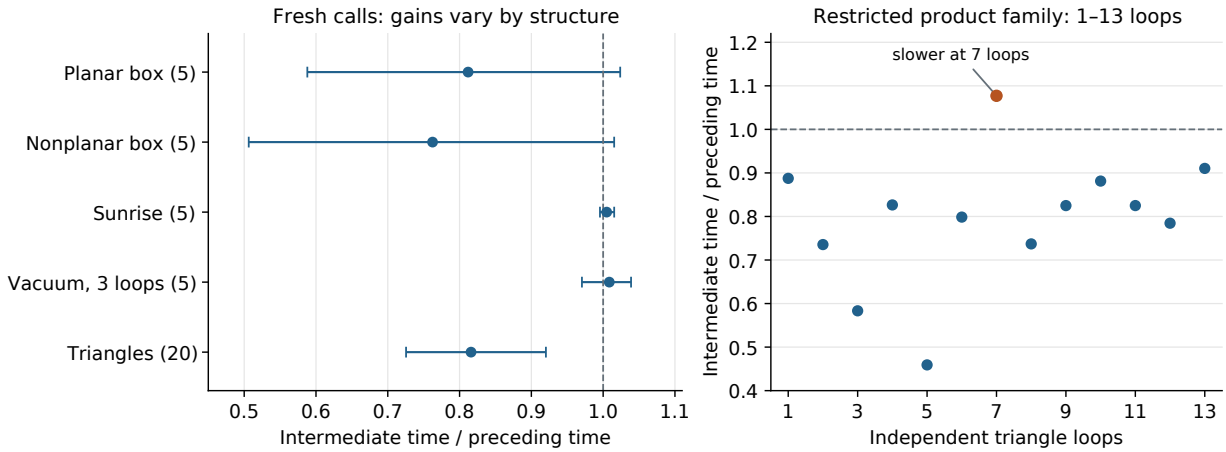


Figure 2: The fresh-call gain depends on the input. Left: ratio of summed median times for each group; each entry uses the same requests in both methods, with request counts in parentheses. Horizontal bars are central 95% paired-request bootstrap intervals. Right: triangle-product ratios at loop orders one through thirteen. There are eight requests at one loop and one at each higher loop order; the right panel therefore shows point estimates without intervals. Values below the dashed line mean less time. These plots show the intermediate method against the preceding method, and use a different timing protocol from Figure 1.

time, and a later compiled row policy had an interval that included equal time. We therefore retain Kira's native row choices.

Table 6: Separate learned-component comparisons from this study. Ratios below one mean less time. The workloads and controls differ, so the ratios cannot be combined into one speed factor. Intervals retain the original study's resampling units, specified below.

Learned component	Control and workload	Ratio	Interval
Earlier seed model	Fixed seed region; 32 box batches	0.963	$[0.937, 0.984]$
Earlier equation model	Native filter; 24 new-index batches	0.933	$[0.928, 0.940]$
Original row model	Native row choices; 84 trials	1.305	—
Compiled row-cost policy	Native choices; geometric-mean estimator	0.982	$[0.959, 1.003]$
Inherited learned choices with reconstruction	Fixed choices, same arithmetic; 22 native requests	0.983	$[0.951, 1.003]$

The seed and reconstruction-era intervals resample requests; the equation interval resamples paired repetitions on fixed requests. The equation timer includes its coefficient checks and excludes initial imports, whereas the final collection has the timing boundary in Section 4. The compiled-row interval retains its original grouped estimator. These differences are another reason to keep the comparisons separate.

The last row tests the inherited learned choices after the arithmetic changes. Its fixed and learned times are 42.175 and 41.448 seconds, respectively, but the interval includes equal time. It therefore does not establish a reliable additional aggregate gain from those learned choices on that sample. Both procedures complete all requests without an unsuccessful internal attempt.

The latest arithmetic selector also fails its validation condition. Its best fitted choice has 46.495 seconds of validation cost, compared with 46.438 seconds for the structure-dependent fixed settings. The required three-percent gain is absent, so the added selector is disabled. Similarly, the nonplanar equation filter takes 16.807 seconds against 9.824 seconds without the filter. It skips 17,808 of 53,430 candidate equation constructions, but all ten filtered attempts fail to reach the prescribed master basis and need recovery. This test demonstrates why equation counts alone cannot establish a time gain.

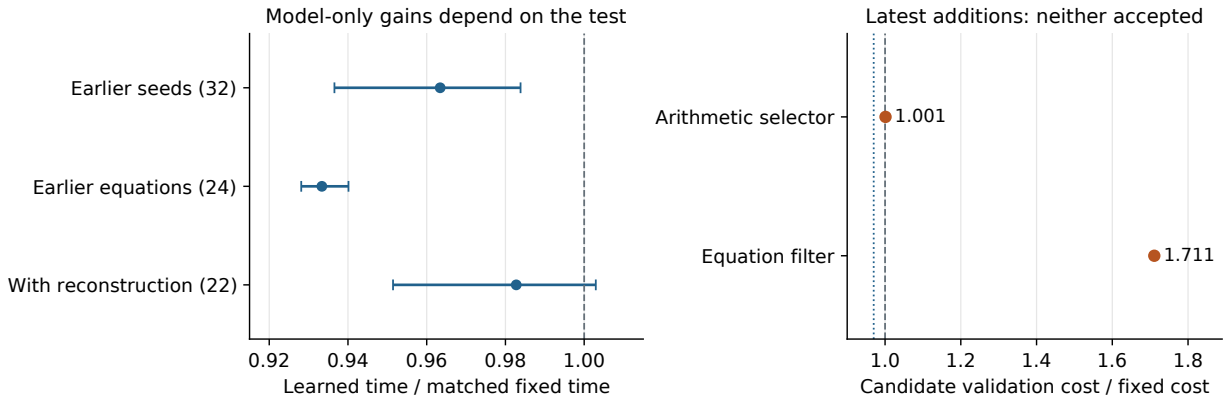


Figure 3: Learned improvements are specific to a tested setting. Left: three model-only time ratios with their recorded resampling intervals and request counts. The seed and reconstruction intervals resample requests; the equation interval resamples paired repetitions. The reconstruction-era interval includes equal time. Right: latest validation costs, normalized to the matched fixed control. The additional arithmetic selector and equation filter remain disabled. The dashed line marks equal cost, and the dotted line marks the arithmetic selector's 0.97 acceptance threshold.

5.4 Earlier combined gains and unsuccessful comparisons

The earlier 75-request experiment measures the effect of adding validated coefficient reconstruction to an application that already has direct vacuum formulas. Summed median time changes from 79.624 to 49.852 seconds, or 37.4% less time. The directly matched comparison across both the vacuum and reconstruction changes gives 81.827 to 49.852 seconds, or 39.1% less time. The latter is a cumulative result on one workload. Its control is an earlier application, not unmodified Kira, and its percentage is separate from the final 93-request result.

An initial parallel version of the final comparison stopped before all three repetitions finished. Of 651 recorded request results, 650 passed and one preceding-method vacuum result failed the prescribed-basis check. After an interrupted attempt, recovery returned 20 masters while the reference used 16. No conversion between those bases was performed, so the test did not establish whether the returned expressions were equivalent. We preserve this failed comparison and exclude its incomplete repetition from the timing estimator. The serial experiment of Table 4 is complete; it does not erase the earlier failure. The notes retain the full interrupted experiment, older timing protocols, and model-training results.

6 Interpretation and limits

The measured gain has several sources. Exact identities remove an elimination problem when their assumptions hold. Finite-field reconstruction changes how Kira carries out the remaining coefficient algebra. Shared calls reduce repeated initialization, and learned choices can reduce the equation system in domains where a matched test supports them. The separate controls in Section 5 show why the combined timing improvement cannot all be attributed to machine learning.

The physically relevant distinction is the structure of the propagators. An independent thirteen-loop product can be reduced by multiplying known component results, and a connected bubble tree can be integrated from its leaves. A graph with additional momentum couplings need not permit either step. Table 1 and the appendix derivations state the domains of the formulas, while Table 3 states the smaller set that received the final complete-time test.

The native benchmarks keep d symbolic and fix the masses and scattering invariants to rational values. Their outputs are exact rational functions of d. The analytic scalar mass formulas also

admit symbolic squared masses in their stated domain. These tests do not establish complete-time performance for arbitrary symbolic multiscale scattering, all numerator powers, or every connected graph through thirteen loops. A finite comparison also cannot prove that the chosen settings minimize time globally.

Recovery deserves particular attention because the cost of a failed restricted system can exceed the work saved by removing equations. The latest equation-filter trial exhibits this effect directly. Its final outputs agree with their references after recovery, but its time is 71.1% greater. Predicting failure from training data can guide a choice, while completion still requires algebraic checks. Exact formulas have a different admission test: their propagator and kinematic conditions must be verified before substitution.

The final application thus supports a concrete use case: requests drawn from the stated analytic domains and native families, with the tested basis and arithmetic settings. Its 8.8% aggregate gain is accompanied by slower subgroups, and the latest model-only evidence remains inconclusive or negative. Wider performance claims require new independent graph structures and complete symbolic-kinematic comparisons with equally tuned controls.

7 Conclusion

We obtain a faster reduction procedure by applying exact identities where their physical conditions hold and selecting the remaining Kira procedure from complete measured costs. The final matched collection takes 41.137 seconds instead of 45.091 seconds, an 8.8% reduction relative to the preceding application. Every result in that completed comparison agrees with its exact reference.

The study separates this combined gain from the effect of learning. Earlier seed and equation models improve their own controlled workloads, while the row model is omitted and the latest arithmetic selector and equation filter remain disabled. Their results show that a learned intervention must reduce complete time after its decision and recovery costs are included.

Exact formulas provide mathematical scope statements: the appendices derive the accepted one-loop, vacuum, triangle, and connected-tree identities. The tests extend these restricted structures through thirteen loops and include native connected families at two and three loops. General high-loop scattering remains a separate problem. The reported method and its limits therefore concern the stated integral domains.

A IBP identities, row operations, and exactness

The starting total-derivative identity follows from integration by parts in a region where the integral converges, followed by dimensional analytic continuation [2, 1]. Write $P_{\boldsymbol{\sigma}} = \prod_i D_i^{-\sigma_i}$ and $\partial_k = \partial/\partial \ell_k$. The product rule gives

$$\partial_k \cdot (v P_{\boldsymbol{\sigma}}) = (\partial_k \cdot v) P_{\boldsymbol{\sigma}} - \sum_i \sigma_i (v \cdot \partial_k D_i) D_i^{-\sigma_i - 1} \prod_{j \neq i} D_j^{-\sigma_j}. \tag{10}$$

For $v = \ell_h$, the first factor is $d\delta_{kh}$; for an external momentum it is zero. Completeness of the inverse-propagator coordinates permits us to write $v \cdot \partial_k D_i = u_i + \sum_j v_{ij} D_j$, where u_i and v_{ij} are coefficients fixed by the propagators and kinematics, with the invariant and mass terms included in u_i. Integrating Eq. (10) therefore gives

$$0 = (\partial_k \cdot v) I_{\boldsymbol{\sigma}} - \sum_i \sigma_i u_i I_{\boldsymbol{\sigma} + \boldsymbol{e}_i} - \sum_{i,j} \sigma_i v_{ij} I_{\boldsymbol{\sigma} + \boldsymbol{e}_i - \boldsymbol{e}_j}. \tag{11}$$

The unit vector $\boldsymbol{e}_i$ raises index i by one. This derivation explains how a seed and an operator generate a row of the reduction system.

For the introductory tadpole example, put $D = \ell^2 - x + i0$ and $v = \ell$. Direct differentiation gives

$$\partial_\ell \cdot (\ell D^{-a}) = dD^{-a} - 2a\ell^2 D^{-a-1}. \tag{12}$$

Using $\ell^2 = D + x$ with the infinitesimal prescription understood, then integrating, gives $(d - 2a)T_a - 2axT_{a+1} = 0$. Solving at $a = 1$ proves Eq. (5). This uses the same analytically continued total-derivative identity as Eq. (3).

Let q be a source row and k a pivot column with $A_{qk} \neq 0$. Solving the identity $\sum_j A_{qj} I_j = 0$ for I_k gives $I_k = -\sum_{j \neq k} (A_{qj}/A_{qk}) I_j$. Subtracting A_{rk}/A_{qk} times that zero identity from row r yields

$$A'_{rj} = A_{rj} - \frac{A_{rk}}{A_{qk}} A_{qj}, \qquad A'_{rk} = 0. \tag{13}$$

Every step preserves the represented relation. Repetition until only prescribed master columns remain gives Eq. (6). Removing seeds or equations preserves the validity of the retained identities, but can prevent this elimination from finishing. We therefore require reduction to the prescribed basis and exact agreement with the reference coefficients. Generic kinematics avoid singular pivot denominators; special kinematic limits require their own treatment.

For independent components, write the loop momenta as disjoint groups and the integrand as $\prod_b f_b$. In a common convergence domain, repeated integration gives $I = \prod_b \int f_b$. If the chosen integer powers have no such domain in d, first vary the propagator powers independently and then continue them to the requested values. Analytic continuation preserves the resulting identity [1]. Substituting each exact component reduction therefore gives

$$I = \prod_b \left(\sum_m r_{bm} M_{bm} \right). \tag{14}$$

Our product output retains this factored form. It avoids expanding the Cartesian product of master terms and uses the same representation in all timed methods.

B The seed regions and equation score

For seed $\boldsymbol{\sigma}$, define excess positive power $D = \sum_i \max(\sigma_i - 1, 0)$, numerator rank $R = \sum_i \max(-\sigma_i, 0)$, and positive sector $P = \{i : \sigma_i > 0\}$. Let h be the number of absent physical lines relative to the largest top sector, and let D_T, R_T be the largest target values. The distance from the target numerator rectangles is defined by

$$q(\boldsymbol{\sigma}, \mathcal{T}) = \min_{\boldsymbol{a} \in \mathcal{T}} \sum_{i \notin P} \max\{0, -\sigma_i - \max(0, -a_i)\}. \tag{15}$$

Projection onto the interval from zero to each target numerator power leaves precisely this positive excess. Summing the distances and minimizing over targets gives Eq. (15). We define the region by

$$\begin{aligned} q &\leq \max\{0, \lfloor \theta_0 + \theta_1 h + \theta_2 D + \theta_6 R \rfloor\}, \\ R &\leq \max\{0, \lfloor R_T + \theta_3 - \theta_4 h \rfloor\}, \qquad D \leq \max\{0, \lfloor D_T + \theta_5 \rfloor\}. \end{aligned} \tag{16}$$

These inequalities specify a search family rather than a sufficiency theorem. Native outer bounds, retained sector corners, and admissible target and master seeds also apply. The previously studied sparse choices use

$$\begin{aligned} \boldsymbol{\theta}_{\text{wide}} &= (2.5, 0, 0, 1.25, 0, 1, -0.25), \\ \boldsymbol{\theta}_{\text{tight}} &= (1, 0, 0, 0, 1, 1, 0), \\ \boldsymbol{\theta}_{\text{medium}} &= (1.5, 0, 0, 0.5, 0, 1, 0). \end{aligned} \tag{17}$$

The retained native model uses these three sparse choices. The rejected replacement uses the wide and tight choices and a third action that retains all seeds within the tight procedure's native outer bounds. The native procedure retains its subsector truncation and recovery rules. The model selects one complete procedure before generating equations.

The separately tested equation classifier has seventeen features: four family and request dimensions; three seed counts; two differences from target power maxima; three sector descriptors; exact target membership and minimum index distance; the operator index; and the two target maxima. Positive labels mark equations retained by a complete reference reduction. Sampling at most 2,000 records of each class y in batch c gives each sampled record weight $N_{cy}/(n_{cy}N_c)$, where N_{cy} and n_{cy} are the native and sampled class counts and N_c is the full batch count. Multiplying this weight by n_{cy} preserves the class fraction N_{cy}/N_c. A leaf with weighted totals W_0, W_1 returns $p = W_1/(W_0 + W_1)$, and a fixed threshold retains equations. The decision-tree construction follows Ref. [14]. This empirical score does not certify sufficiency or probability calibration; the native completion check remains necessary.

C Measured-cost learning

Let i label a training request and a an allowed arithmetic procedure. In the latest study, C_{ia} is the measured complete time, including recovery, and $b_i > 0$ is the time for the preceding procedure on the same request. Define the relative training target by

$$y_{ia} = C_{ia}/b_i. \tag{18}$$

There are five actions, so the latest model predicts the vector $\boldsymbol{z}_i = (y_{i1}, \ldots, y_{i5})$. These quantities are training definitions. The three-loop structure uses the preceding procedure by an explicit guard and is excluded from the 36-request fit. The earlier reconstruction study used six actions and an estimated recovery penalty; its separate objective and outcomes remain in the notes.

The ensemble has 32 extremely randomized regression trees [16]. At a terminal branch, or leaf, let S be the set of training requests that reach it and let μ be the predicted value of one output component z. We use the loss $\sum_{i\in S} b_i(z_i - \mu)^2$, which defines the baseline-time weighting. Differentiating gives

$$0 = 2\sum_{i\in S} b_i(\mu - z_i), \qquad \widehat{z}(S) = \frac{\sum_{i\in S} b_i z_i}{\sum_{i\in S} b_i}. \tag{19}$$

Substituting the relative cost from Eq. (18) gives

$$\widehat{y}_a(S) = \frac{\sum_{i\in S} C_{ia}}{\sum_{i\in S} b_i}, \qquad \arg\min_a \widehat{y}_a(S) = \arg\min_a \sum_{i\in S} C_{ia}. \tag{20}$$

The second equality follows because the denominator is positive and independent of the action. Thus an individual leaf ranks actions by their aggregate measured training cost. The ensemble averages its 32 tree predictions. This construction does not establish a minimum cost on unseen requests.

For the fixed action a_0 selected for the request's propagator structure, an alternative a_* is proposed only if

$$\widehat{y}_{a_*} < (1 - \eta)\widehat{y}_{a_0}. \tag{21}$$

The margin η is a defined acceptance parameter. Validation compares $\eta = 0, 0.05, 0.10$, tree depths 2, 4, and 6, and minimum leaf populations 2 and 4. The complete selected policy must cost less than 0.97 times its structure-dependent fixed control and select no failed validation action. The latest policy does not pass this condition. The final application therefore uses full reconstruction with eight evaluation points grouped together for the planar box, nonplanar box, and sunrise structures, and the preceding procedure for the three-loop vacuum structure.

To specify the features, define target excess power $D = \sum_j \max(a_j - 1, 0)$, numerator rank $R = \sum_j \max(-a_j, 0)$, and line count $P_T = \sum_j \mathbf{1}_{a_j>0}$. A line is incident on a loop momentum when that momentum occurs in its inverse propagator. The 29 features are listed in Table 7. Every feature follows from the supplied indices and propagator definitions. The ordering within

Table 7: Inputs to the algebra-cost model. Means and extrema over targets use the complete requested batch.

Positions	Quantities
1–6	Number of loops, external momenta, integral coordinates, physical propagators, symbolic external invariants, and $\log(1 + n_T)$ for n_T targets
7–13	Maximum and mean D, maximum and mean R, minimum and maximum P_T, and number of distinct sectors
14–18	Number of massive coordinates; number of physical lines containing several loops; maximum loop incidence per physical line; maximum physical-line incidence per loop; mean external-momentum incidence per coordinate
19–22	Minimum D, minimum R, mean P_T, and maximum absolute index
23–25	Mean excess power on massive coordinates; mean numerator rank weighted by loop incidence; mean excess power weighted by loop incidence
26–29	Maximum sum of positive powers; maximum number of absent physical lines; maximum numerator and denominator bit lengths among rational masses and scalar products

positions 7–10 is maximum D, maximum R, mean D, mean R. The remaining entries follow the listed order. The bit lengths describe the supplied rational numbers and do not encode a family identifier.

D Physical test inputs

All profiles below define squared quantities in a common fixed unit. The zero-invariant families use shifts $(0, p, q, r)$ and masses $(x, y, 0, 0)$, with

$$\begin{aligned} &p^2 = 0, \quad p \cdot q = -h - 1, \quad p \cdot r = -h - 2, \\ &q^2 = -2h - 7, \quad q \cdot r = -h - 3, \quad r^2 = -3h - 13. \end{aligned} \tag{22}$$

The unequal case has $(x, y) = (h+3, h+11)$, the equal case $(h+3, h+3)$, the mixed case $(h+3, 0)$, and the massless case $(0, 0)$. Targets have the form $(1 + \delta_1, 1 + \delta_2, -b_1, -b_2)$, where $\delta_1 + \delta_2$ is the excess positive power and $b_1 + b_2$ is numerator rank.

Ordinary tensor families have P physical lines, one auxiliary coordinate, shifts $q_1 = 0$, $q_{i+1} = p_i$, and

$$m_i^2 = h+11i^2 \ (i \le P), \quad m_{P+1}^2 = 0, \quad p_i^2 = -(10h+17i), \quad p_i \cdot p_j = -(h+i+2j-2) \ (i < j). \tag{23}$$

Appendix J specifies the current profiles, targets, and separate vacuum grids.

For the native planar box, let p_1, p_2, p_3 be massless, $2p_1 \cdot p_2 = s$, $2p_1 \cdot p_3 = t$, and $2p_2 \cdot p_3 = -s-t$. The seven physical line momenta are

$$(\ell_1, \ell_1 - p_1, \ell_1 - p_1 - p_2, \ell_2, \ell_2 - p_1 - p_2, \ell_2 - p_1 - p_2 - p_3, \ell_1 - \ell_2), \tag{24}$$

and the auxiliary momenta are $(\ell_1 - p_3, \ell_2 - p_1)$. The nonplanar routing follows Ref. [4]:

$$\begin{aligned} (Q_1, \ldots, Q_9) = (&\ell_1, \ell_2, p_1 - \ell_1, p_2 - \ell_2, p_1 + p_2 - p_3 - \ell_1, \\ &p_1 - \ell_1 + \ell_2, p_1 - p_3 - \ell_1 + \ell_2, \ell_1 - p_3, \ell_2 - p_1 - p_2). \end{aligned} \tag{25}$$

The first seven entries are physical and all masses vanish. Its convention gives $p_i^2 = (p_1 + p_2 - p_3)^2 = 0$, $s = (p_1 + p_2)^2$, and $t = (p_1 - p_3)^2$. Expanding the squares gives $2p_1 \cdot p_3 = -t$ and

$2p_2 \cdot p_3 = s + t$. The low-rank profiles use $(s, t) = (-1009 - 12h, -1021 - 14h)$ for the planar graph and $(-113-18h, -127-10h)$ for the nonplanar graph. Appendix K gives the independently chosen high-rank invariants.

Every high-rank box target is $(1, 1, 1, 1, 1, 1, 1, -u, -v)$ with $u, v \geq 0$ and $1 \leq u + v \leq 4$. The current split uses the profiles and batch counts in Appendix K. Both graph types occur in all splits, so the test measures new batches and kinematic points within these structures.

E Time ratios and uncertainty

For the final shared-call comparison, let U_{mr} be elapsed time for the entire 93-request collection, method m, and repetition r. We define

$$\widetilde{U}_m = \operatorname{median}_{r=1,2,3} U_{mr}, \qquad \rho_{\text{collection}} = \frac{\widetilde{U}_{\text{final}}}{\widetilde{U}_{\text{preceding}}}, \qquad g = 1 - \rho_{\text{collection}}. \tag{26}$$

The saving g follows by dividing the time difference $\widetilde{U}_{\text{preceding}} - \widetilde{U}_{\text{final}}$ by the preceding time. The observed range is $[\min_r U_{mr}, \max_r U_{mr}]$. It describes the three measurements and is not a statistical confidence interval. Internal per-request timers exclude some initialization and collection overhead, so summing their medians need not give $\widetilde{U}_m$.

Let T_{imr} be complete elapsed time for request i, method m, and repetition r. For the set $\mathcal{C}$ completed by both methods in all repetitions, define

$$S_m = \sum_{i \in \mathcal{C}} \operatorname{median}_{r=1,2,3} T_{imr}, \qquad \rho = \frac{S_{\text{new}}}{S_{\text{control}}}, \qquad \text{fractional saving} = 1 - \rho. \tag{27}$$

These definitions compare identical completed requests. Dividing both times by the control time gives ρ, and subtracting gives the fractional saving. Completion counts remain separate because an unfinished reduction has no verified finite speedup. To estimate variation across requests, we draw $|\mathcal{C}|$ paired request indices with replacement and recompute ρ. We report the central 95% of 5,000 such ratios, using the bootstrap construction of Ref. [17]. This interval describes the sampled request collection and does not establish performance for every integral family.

For the historical compiled-row result, the estimator is the geometric mean of paired ratios, $\rho_G = \exp[n^{-1} \sum_{i=1}^{n} \log(t_i^{\text{learned}} / t_i^{\text{native}})]$, where n is the number of matched units in that study. This is the nth root of their product, obtained by applying the logarithm to that product. It weights relative changes equally, unlike the total-time ratio in Eq. (27); the notes preserve the original grouping and resampling procedure.

F Exact reduction of polynomial numerators

F.1 Resolving the loop momentum

One-loop tensor reduction expresses loop-momentum numerators through scalar integrals. Established methods include tensor decompositions and generating functions for arbitrary numerator and denominator powers [7, 8]. We derive the particular identities used in our extension here. The algebra uses rational kinematic values and keeps d symbolic. It applies to a one-loop sector and to independent products of such sectors. Applying it inside a coupled graph would require coefficients that depend on the other loop momenta, which this implementation does not support.

Write $D_i = (\ell - q_i)^2 - m_i^2$ and let $S = \{i : a_i > 0\}$ contain the denominator positions. The q_i in this appendix are fixed external shifts; the infinitesimal prescription is understood. Choose

$r \in S$, set $k = \ell - q_r$, and define $R_i = q_i - q_r$. The vectors R_i with $i \in S \setminus \{r\}$ span the external directions seen by the denominators. Their Gram matrix and their contractions with k are

$$G_{ij} = R_i \cdot R_j, \qquad u_i = k \cdot R_i = \frac{D_r - D_i + m_r^2 - m_i^2 + R_i^2}{2}. \tag{28}$$

The second equality follows by expanding $D_i = (k - R_i)^2 - m_i^2$ and subtracting it from $D_r = k^2 - m_r^2$. Assume that G is invertible, and split k into its projection onto this span and its perpendicular part:

$$k_\parallel^\mu = \sum_{i,j \in S\setminus\{r\}} R_i^\mu (G^{-1})_{ij} u_j, \qquad k_\perp = k - k_\parallel. \tag{29}$$

Contracting with each R_i gives $k_\perp \cdot R_i = 0$. Contracting the parallel part with itself gives $k_\parallel^2 = u^T G^{-1} u$, so

$$T(D) := k_\perp^2 = D_r + m_r^2 - u^T G^{-1} u. \tag{30}$$

Empty matrices and sums give zero when S has one element.

An inverse propagator D_j with $a_j < 0$ appears in the numerator and can be resolved into the same components. Define $v_{ji} = R_j \cdot R_i$, $R_{j,\perp} = R_j - \sum_{i,h} R_i (G^{-1})_{ih} v_{jh}$, and $z_j = k_\perp \cdot R_{j,\perp}$. Direct substitution gives

$$D_j = B_j(D) - 2z_j, \tag{31}$$

$$B_j(D) = D_r + m_r^2 + R_j^2 - m_j^2 - 2 v_j^T G^{-1} u, \tag{32}$$

$$H_{jh} := R_{j,\perp} \cdot R_{h,\perp} = R_j \cdot R_h - v_j^T G^{-1} v_h. \tag{33}$$

Thus the remaining task is to average products of the z_j over perpendicular directions. All positive-power denominators depend on $k_\perp$ only through $T(D)$, so this angular average leaves the denominator unchanged.

F.2 Angular moments and the finite numerator expansion

Let $\nu = d - |S| + 1$ be the dimension perpendicular to the denominator span. Rotational symmetry makes odd angular moments vanish. For $2h$ perpendicular components, the invariant symmetric tensor is the sum of all pairings of metric tensors, multiplied by a coefficient $C_h T^h$. Tracing two indices gives

$$C_h(\nu + 2h - 2) = C_{h-1}, \qquad C_0 = 1, \qquad C_h = \frac{1}{\prod_{b=0}^{h-1}(\nu + 2b)}. \tag{34}$$

The trace factor contains ν from the pairing of the traced indices and $2h - 2$ from their pairings with the other indices. Contracting the tensor with the external perpendicular vectors therefore gives

$$\left\langle \prod_{t=1}^{2h} z_{j_t} \right\rangle = \frac{T(D)^h}{\prod_{b=0}^{h-1}(\nu + 2b)} \sum_{\mathcal{P}} \prod_{(s,t)\in\mathcal{P}} H_{j_s j_t}. \tag{35}$$

Here $\mathcal{P}$ runs over the distinct pairings of the $2h$ positions, including their multiplicities when an index repeats. In particular,

$$\langle z_i z_j \rangle = \frac{T H_{ij}}{\nu}, \qquad \langle z_i z_j z_k z_l \rangle = \frac{T^2 (H_{ij}H_{kl} + H_{ik}H_{jl} + H_{il}H_{jk})}{\nu(\nu + 2)}. \tag{36}$$

One can first derive these relations after a Wick rotation in an integer dimension, where the angular average is well defined, and then continue the rational tensor coefficients in d. This uses the dimensional-regularization convention of Eq. (1) [1]. Poles in d remain exact rational poles.

For $b_j = -a_j > 0$, expand $\prod_{j:a_j<0}(B_j - 2z_j)^{b_j}$ and insert Eq. (35). The result is a finite polynomial in the positive-sector denominators:

$$\left\langle \prod_{j:a_j<0} D_j^{b_j} \right\rangle = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) \prod_{i\in S} D_i^{t_i}, \qquad I_{\boldsymbol{a}} = \sum_{\boldsymbol{t}} c_{\boldsymbol{t}}(d) I_{\boldsymbol{a}_S - \boldsymbol{t}}. \tag{37}$$

The vector $\boldsymbol{a}_S$ equals $\boldsymbol{a}$ on S and vanishes elsewhere; the t_i are nonnegative and vanish outside S. The coefficients also depend rationally on the fixed kinematics. Each power $D_i^{t_i}$ cancels the corresponding denominator power. If a cancellation creates a new negative index, that line leaves the positive sector. Repeating the angular step therefore decreases the positive-sector size whenever a numerator remains. A sector with no denominators is scaleless and vanishes in dimensional regularization [1]. The process terminates in scalar sectors with nonnegative indices.

F.3 Lowering powers before removing the numerator

A second exact route starts with IBP. Define $A = \sum_j a_j$ and $Y_{ij} = m_i^2 + m_j^2 - (q_i - q_j)^2$. Expanding the total derivative with vector $\ell - q_i$ gives

$$0 = dI_{\boldsymbol{a}} - \sum_j a_j \int \frac{d^d\ell}{i\pi^{d/2}} \frac{2(\ell - q_i)\cdot(\ell - q_j)}{D_j} \prod_h D_h^{-a_h}, \tag{38}$$

$$2(\ell - q_i)\cdot(\ell - q_j) = D_i + D_j + Y_{ij}, \tag{39}$$

$$0 = (d - A)I_{\boldsymbol{a}} - \sum_j a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_j a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{40}$$

Here $\boldsymbol{e}_j$ is the unit index vector. This derivation holds for signed integer indices, so it includes numerator terms. Let $N_- = \{j : a_j < 0\}$. Moving the negative-index terms to the right gives, for $i \in S$,

$$\sum_{j\in S} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j} = F_i(\boldsymbol{a}), \tag{41}$$

$$F_i(\boldsymbol{a}) = (d - A)I_{\boldsymbol{a}} - \sum_{j:a_j\neq 0} a_j I_{\boldsymbol{a}+\boldsymbol{e}_j-\boldsymbol{e}_i} - \sum_{j\in N_-} a_j Y_{ij} I_{\boldsymbol{a}+\boldsymbol{e}_j}. \tag{42}$$

For target $\boldsymbol{b}$ with $b_k > 1$, choose $\boldsymbol{a} = \boldsymbol{b} - \boldsymbol{e}_k$. If Y_S is invertible, multiply this system by its inverse to obtain

$$I_{\boldsymbol{b}} = \frac{1}{a_k} \sum_{i\in S} (Y_S^{-1})_{ki} F_i(\boldsymbol{a}). \tag{43}$$

Every integral on the right has a smaller sum of positive indices than $\boldsymbol{b}$, and the numerator rank does not increase. When the positive powers reach one, the angular step removes the remaining numerator. With no negative indices, Eq. (43) reduces to the scalar Cayley recurrence derived earlier. These identities give a finite reduction to unit-power scalar sectors. They produce a spanning set; special kinematics can introduce further relations between its elements.

F.4 Keeping the dimension poles in factored form

The preceding identities restrict the form of every dimension-dependent denominator. The inverse Gram and Cayley matrices have rational entries at the chosen kinematic point, and Eq. (43) introduces only polynomial dependence on d. Equation (35) introduces products of linear factors $d + j$, where j is an integer. By induction through the recurrence, each coefficient therefore has the form

$$r(d) = \frac{P(d)}{\prod_j (d + j)^{n_j}}, \qquad P(d) \in \mathbb{Q}[d], \quad n_j \in \mathbb{Z}_{\geq 0}. \tag{44}$$

The finite products contain only factors introduced by the angular moments. For addition, take Q to be the least common multiple of two such denominators Q_1 and Q_2. Multiplying each summand by its missing factors gives

$$\frac{P_1}{Q_1} + \frac{P_2}{Q_2} = \frac{P_1(Q/Q_1) + P_2(Q/Q_2)}{Q}. \tag{45}$$

The exponent of $d+j$ in Q is the larger of its two exponents. Multiplication adds these exponents. A factor $d+j$ cancels exactly when $P(-j) = 0$, because division with remainder gives $P(d) = (d+j)\widetilde{P}(d) + P(-j)$. We repeat this cancellation for each retained factor. These operations preserve the exact rational function and require only rational polynomial coefficients and integer factor multiplicities. They avoid general rational-function simplification in the new path. The earlier arithmetic checks compare these coefficients with a general rational-function implementation and native Kira reductions. The current application retains the fixed numerator-first recurrence; its new learned decision concerns native seed regions.

G Exact reduction at a zero two-point invariant

A singular Gram matrix prevents the projection used by the generic tensor recurrence. For a two-point sector with $p^2 = 0$, Feynman parameters give a direct reduction to tadpoles. Degenerate one-loop kinematics require such separate identities [7]. We derive the identity used in this study below. It applies to a polynomial numerator and two positive denominator powers; it does not assume that the two masses differ.

G.1 Scalar integrals and mass derivatives

Let x and y denote squared masses, and define

$$T(x) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{\ell^2 - x + i0}, \qquad B_{ab}(x,y) = \int \frac{d^d\ell}{i\pi^{d/2}} \frac{1}{(\ell^2 - x + i0)^a[(\ell - p)^2 - y + i0]^b}\bigg|_{p^2=0}. \tag{46}$$

Here a, b are positive integers. Dimensional regularization sets $T(0) = 0$. With the normalization in Eq. (1), Wick rotation and the radial beta integral give [1]

$$T(x) = -\Gamma(1 - d/2)(x - i0)^{d/2-1}. \tag{47}$$

For $a = b = 1$, combine the denominators with a parameter z and shift $\ell = k + (1 - z)p$. The combined squared mass is $M^2(z) = zx + (1 - z)y$, because $p^2 = 0$. Integrating over k and then z gives

$$\begin{aligned} B_{11}(x,y) &= \Gamma(2 - d/2) \int_0^1 [zx + (1 - z)y - i0]^{d/2-2}\, dz \\ &= \frac{T(x) - T(y)}{x - y}, \qquad x \neq y. \end{aligned} \tag{48}$$

The second equality follows from the elementary antiderivative of a power and $\Gamma(2 - d/2) = (1 - d/2)\Gamma(1 - d/2)$. Differentiating a propagator with respect to its squared mass raises its power. Therefore

$$B_{ab}(x,y) = \frac{\partial_x^{a-1}\partial_y^{b-1} B_{11}(x,y)}{(a-1)!(b-1)!}. \tag{49}$$

This operation preserves a rational combination of $T(x)$ and $T(y)$ because $\partial_x T(x) = (d/2 - 1)T(x)/x$. We take the derivatives before imposing relations between the masses. If $x = y \neq 0$, the combined mass is independent of z, and the beta integral instead gives

$$B_{ab}(x,x) = T_{a+b}(x), \qquad T_n(x) = \frac{\prod_{j=1}^{n-1}(d/2 - j)}{(n-1)!\, x^{n-1}} T(x). \tag{50}$$

The formula for T_n also follows by differentiating Eq. (47). These identities support independent symbolic squared masses as well as rational values. A relation involving distinct symbolic masses is a generic rational identity; equal masses use Eq. (50) directly.

G.2 Polynomial numerators

Let $P(\ell)$ be a polynomial numerator. Combining the two denominators gives

$$B_{ab}[P] = \frac{\Gamma(A)}{\Gamma(a)\Gamma(b)} \int_0^1 dz\, z^{a-1}(1-z)^{b-1} \int \frac{d^dk}{i\pi^{d/2}} \frac{P(k+(1-z)p)}{[k^2 - M^2(z) + i0]^A}, \quad A = a+b. \tag{51}$$

For an auxiliary inverse propagator $(\ell - q_j)^2 - m_j^2$, put $w_j = (1-z)p - q_j$. Its numerator factor becomes $k^2 + 2k\cdot w_j + w_j^2 - m_j^2$. Both $w_i \cdot w_j$ and $w_j^2 - m_j^2$ are polynomials of degree at most one in z when $p^2 = 0$.

Rotational covariance gives zero for odd moments of k. Tracing two indices of an even moment determines its normalization recursively: if its pairing coefficient is C_h, then $C_h(d+2h-2) = C_{h-1}$ and $C_0 = 1$. Consequently,

$$\left\langle \prod_{j=1}^{2h} (k\cdot w_j) \right\rangle = \frac{(k^2)^h}{\prod_{r=0}^{h-1}(d+2r)} \sum_{\mathcal{P}} \prod_{(i,j)\in\mathcal{P}} w_i \cdot w_j, \tag{52}$$

where $\mathcal{P}$ runs over pairings of the $2h$ positions. The trace recurrence follows because one pairing contributes d and the remaining contractions contribute $2h-2$. This also derives the full-dimensional form of the angular identity used in Appendix F.

After the finite numerator expansion and angular average, define $c_{ht}(d)$ by

$$\langle P(k+(1-z)p)\rangle = \sum_{h,t} c_{ht}(d)(k^2)^h z^t. \tag{53}$$

These coefficients also depend on the fixed external scalar products and numerator masses. They contain no loop momentum. To integrate each term, Wick rotate the radial integral, set $u = k_E^2/M^2$, and use $\int_0^\infty u^{\alpha-1}(1+u)^{-A}du = \Gamma(\alpha)\Gamma(A-\alpha)/\Gamma(A)$ with $\alpha = d/2 + h$. To obtain this beta identity, multiply the defining Gamma integrals for $\Gamma(\alpha)$ and $\Gamma(A-\alpha)$, and set their variables to rz and $r(1-z)$. The r integral is $\Gamma(A)$; the remaining integral is converted by $u = z/(1-z)$ to the stated expression. This gives

$$J_{Ah}(M^2) = \int \frac{d^dk}{i\pi^{d/2}} \frac{(k^2)^h}{(k^2 - M^2 + i0)^A} = (-1)^{A-h} \frac{\Gamma(d/2+h)\Gamma(A-h-d/2)}{\Gamma(d/2)\Gamma(A)} (M^2 - i0)^{d/2+h-A}. \tag{54}$$

The initial convergence region defines the integrals; analytic continuation gives the dimensional-regularization identity outside that region [1].

For $x \neq y$, set $\Delta = x - y$, $\rho = h - A + 1$, and $\lambda = d/2 - 1$. Expand $(1-z)^{b-1}$ and then substitute $u = y + \Delta z$. For an integer $k \geq 0$,

$$\int_0^1 z^k (y+\Delta z)^{\lambda+\rho}\, dz = \frac{1}{\Delta^{k+1}} \sum_{j=0}^{k} \binom{k}{j} (-y)^{k-j} \frac{x^{\lambda+\rho+j+1} - y^{\lambda+\rho+j+1}}{\lambda+\rho+j+1}. \tag{55}$$

This follows by expanding $(u-y)^k$ and integrating each power of u. Dividing by the tadpole normalization in Eq. (47), define

$$R_h = \frac{(-1)^{A-h+1}(d/2)_h (1-d/2)_{A-h-1}}{(a-1)!(b-1)!}, \qquad (z)_n = \frac{\Gamma(z+n)}{\Gamma(z)}. \tag{56}$$

For a negative integer n, the last ratio means the inverse finite product $1/\prod_{r=1}^{-n}(z-r)$. Equations (51)–(55) now give the explicit rational reduction

$$\begin{aligned} B_{ab}[P] &= \sum_{h,t} c_{ht}(d) R_h \sum_{v=0}^{b-1} (-1)^v \binom{b-1}{v} \\ &\quad \times \sum_{j=0}^{k} \binom{k}{j} \frac{(-y)^{k-j}}{\Delta^{k+1}} \frac{x^{\rho+j+1} T(x) - y^{\rho+j+1} T(y)}{\lambda+\rho+j+1}, \\ &\quad k = a-1+t+v. \end{aligned} \tag{57}$$

If one mass is zero, its endpoint vanishes in a convergence region and hence by analytic continuation; the implementation omits that scaleless endpoint before evaluating powers. If both masses vanish, the sector is scaleless. For equal nonzero masses, use the beta integral directly:

$$B_{ab}[P] = \sum_{h,t} c_{ht}(d) R_h \frac{\Gamma(a+t)\Gamma(b)}{\Gamma(A+t)} x^{h-A+1} T(x). \tag{58}$$

Thus the equal-mass rule has no division by $x-y$. A sector with one positive denominator uses Eq. (54) and the same angular expansion without a Feynman parameter.

At rational kinematics, the coefficients have polynomial numerators in d and products of integer-shift linear factors $d+j$ in their denominators. The finite products in Eqs. (52) and (56), and the linear denominator in Eq. (55), prove this statement. We therefore use the exact factored arithmetic of Appendix F. The implemented tensor domain has at most five coordinates, numerator rank at most four, and at most four extra positive powers. Symbolic masses currently use the scalar derivative identity. Inputs outside these conditions continue through the existing exact rules and Kira.

H Connected one-scale tree reductions

We derive the previously retained tree rules to state precisely which high-loop structures they cover. Let $\mathcal{T}$ be a rooted tree with loop momenta $\ell_0, \dots, \ell_{L-1}$ and parent $\pi(j)$ for every nonroot vertex. A massless two-point family is

$$J_{\mathcal{T}} = \int \prod_{j=0}^{L-1} \frac{d^d \ell_j}{i\pi^{d/2}} \frac{1}{(\ell_0^2)^{a_0}[(\ell_0-p)^2]^c} \prod_{j=1}^{L-1} \frac{1}{(\ell_j^2)^{a_j}[(\ell_j-\ell_{\pi(j)})^2]^{v_j}}, \qquad p^2 = s \neq 0. \tag{59}$$

The massive vacuum version replaces the two root factors by $(\ell_0^2-m^2)^{-c}$ and has no external momentum. Each edge power v_j, root power c, and leaf self power a_j is positive. Internal self powers can be nonpositive. Auxiliary powers vanish. The implementation admits $L \le 13$ and absolute physical powers at most sixteen. The derivation itself applies to finite trees with meromorphically defined integrals.

Temporarily use negative massless denominators. Feynman parametrization of a massless bubble, translation of its loop momentum, and a radial beta integral give

$$\begin{aligned} G(u,v) &= \frac{\Gamma(u+v-d/2)}{\Gamma(u)\Gamma(v)} \int_0^1 z^{d/2-v-1}(1-z)^{d/2-u-1}\, dz \\ &= \frac{\Gamma(u+v-d/2)\Gamma(d/2-u)\Gamma(d/2-v)}{\Gamma(u)\Gamma(v)\Gamma(d-u-v)}. \end{aligned} \tag{60}$$

The remaining scale factor is $(-P^2-i0)^{d/2-u-v}$ for bubble momentum P. The beta identity used here follows by the same radial and parameter integrations as Appendix G; analytic continuation supplies the dimensionally regularized relation [1]. Recursive subintegral integration has established applications to massless propagator integrals [18].

For each nonroot vertex, define

$$u_j = a_j + \sum_{h:\pi(h)=j} \sigma_h, \qquad \sigma_j = u_j + v_j - d/2. \tag{61}$$

Integrating a leaf gives $G(u_j, v_j)$ and transfers power σ_j to its parent. Repeating from the leaves to the root proves the product rule by induction. In the two-point case, set $\alpha = a_0 + \sum_{h:\pi(h)=0} \sigma_h$; its dimensionless coefficient is

$$C_{\mathcal{T}} = G(\alpha, c) \prod_{j=1}^{L-1} G(u_j, v_j). \tag{62}$$

The total physical power $A = \sum_j a_j + \sum_{j>0} v_j + c$ gives scale power $(-s - i0)^{Ld/2-A}$. Let $M_{\mathcal{T}}$ set all physical powers to one and let C_0 be its Gamma product. Restoring the original propagator signs and dividing by this master gives

$$J_{\mathcal{T}} = s^{-(A-2L)} \frac{C_{\mathcal{T}}}{C_0} M_{\mathcal{T}}. \tag{63}$$

The sign $(-1)^{A-2L}$ from the propagator convention cancels that from $(-s)^{-(A-2L)}$.

For a massive vacuum root, put $\sigma_0 = \sum_{h:\pi(h)=0} \sigma_h$. Wick rotation and the substitution $x = m^2 t$ give

$$\frac{1}{\Gamma(d/2)} \int_0^\infty \frac{x^{d/2-\sigma_0-1}}{(x+m^2)^c}\, dx = (m^2)^{d/2-\sigma_0-c} \frac{\Gamma(d/2-\sigma_0)\Gamma(\sigma_0+c-d/2)}{\Gamma(d/2)\Gamma(c)}. \tag{64}$$

Multiply the dimensionless Gamma ratio by the nonroot bubble factors to define $C^V_{\mathcal{T}}$. There are $P = 2L-1$ physical lines and $A = c + \sum_{j>0}(a_j + v_j)$. Dividing by the unit-power vacuum master $M^V_{\mathcal{T}}$ yields

$$V_{\mathcal{T}} = (-m^2)^{-(A-P)} \frac{C^V_{\mathcal{T}}}{C^V_0} M^V_{\mathcal{T}}. \tag{65}$$

In either family, a subtree with n_j vertices has $u_j = U_j - (n_j - 1)d/2$ with integer U_j. This follows from Eq. (61) by induction. Target and master Gamma arguments therefore have equal slopes in d and integer differences in their constant terms. Repeated use of $\Gamma(z+1) = z\Gamma(z)$ gives

$$\frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{k=0}^{n-1}(z+k), & n > 0, \\ 1, & n = 0, \\ [\prod_{k=1}^{-n}(z-k)]^{-1}, & n < 0. \end{cases} \tag{66}$$

Thus the target-to-master coefficients are rational in d and the one scale. Cancelling equal Gamma factors before expanding these finite products preserves the identity and avoids repeated symbolic work. This proves the reductions for the stated trees; additional coupled propagators or auxiliary numerators require another reduction.

I Direct evaluation of vacuum factors

We first identify a case that requires no master integral. If all masses vanish, there is no external momentum, and every active inverse propagator is the square of a homogeneous linear combination of loop momenta, the integrand has no dimensional scale. After Wick rotation, let R be the common radial coordinate for all loop momenta. The measure and denominators give radial power $R^{\omega-1}$ with $\omega = Ld - 2\sum_j a_j$. Split the radial integral at an arbitrary $\mu > 0$ and continue its two pieces separately:

$$\int_0^\mu R^{\omega-1}\, dR + \int_\mu^\infty R^{\omega-1}\, dR = \frac{\mu^\omega}{\omega} - \frac{\mu^\omega}{\omega} = 0. \tag{67}$$

The two pieces initially converge on opposite half-planes and their meromorphic continuations cancel. Angular singularities can be treated by analytic regularization before the same scaling argument. This is the standard scaleless-integral prescription in dimensional regularization [1]. It gives the combined integral as zero; separating ultraviolet and infrared poles requires retaining the two regulated contributions separately. The implementation checks the masses and momentum homogeneity, rejects a zero denominator with positive power, and accepts up to thirteen loops and 128 coordinates. It excludes cut denominators and symbolic-index inputs. Couplings between loop momenta are permitted for this zero-integral rule.

Consider a loop momentum k that occurs only in denominators $k^2 - x_j + i0$, where x_j is a squared mass. There is no external momentum in this component. Denominator powers can be positive, zero, or negative. Equal masses first combine by addition of their powers. With $z = k^2$, let A_x be the resulting integer power at each distinct squared mass x. The integrand is the rational function

$$F(z) = \prod_x (z - x)^{-A_x}. \tag{68}$$

For a mass with $A_x > 0$, multiply by $(z-x)^{A_x}$ and expand the remaining analytic factor at $z = x$. Its first A_x Taylor coefficients give the principal part at that pole. Subtracting every principal part leaves a rational function without a finite pole, which is a polynomial $P(z)$. Thus

$$F(z) = P(z) + \sum_{x:A_x>0} \sum_{r=1}^{A_x} \frac{c_{xr}}{(z-x)^r}, \tag{69}$$

$$c_{xr} = \frac{1}{(A_x - r)!} \frac{d^{A_x - r}}{dz^{A_x - r}} \prod_{y \neq x} (z-y)^{-A_y} \bigg|_{z=x}. \tag{70}$$

This derivation also covers negative powers, since they supply polynomial factors in the product. For distinct rational masses, the coefficients are rational numbers. The implementation obtains them by multiplying truncated Taylor series, so it avoids general symbolic differentiation. A factor with exponent $-A_y$ has coefficient

$$[t^n](x - y + t)^{-A_y} = \binom{-A_y}{n} (x-y)^{-A_y - n}. \tag{71}$$

The binomial series proves Eq. (71); only $0 \le n < A_x$ is needed in Eq. (70). Equal masses must be combined before this expansion, which avoids division by a vanishing mass difference.

Polynomial terms in k and massless vacuum integrals have no scale and vanish in dimensional regularization [1]. For a nonzero squared mass x, define the normalized tadpole

$$T_r(x) = \int \frac{d^d k}{i\pi^{d/2}} (k^2 - x + i0)^{-r}. \tag{72}$$

To evaluate it, first take $x > 0$ and a convergent Euclidean domain. The Schwinger representation follows from the defining Gamma integral after the change of variable $u = \alpha(q^2 + x)$:

$$(q^2 + x)^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty \alpha^{r-1} e^{-\alpha(q^2+x)}\, d\alpha. \tag{73}$$

For $\alpha > 0$, let $G_1 = \int_{-\infty}^{\infty} e^{-\alpha q^2}\, dq$. Squaring this integral and using polar coordinates in two dimensions gives $G_1^2 = 2\pi \int_0^\infty r e^{-\alpha r^2} dr = \pi/\alpha$. Taking its positive square root and multiplying independent Cartesian integrals gives $\int d^d q\, e^{-\alpha q^2} = \pi^{d/2} \alpha^{-d/2}$ for positive integer d. Its analytic continuation supplies the expression used in dimensional regularization. Inserting this in Eq. (73) and setting $u = \alpha x$ gives $x^{d/2-r}\Gamma(r - d/2)/\Gamma(r)$. Wick rotation supplies $(-1)^r$ for the denominator convention in Eq. (72). Analytic continuation therefore gives

$$T_r(x) = (-1)^r \frac{\Gamma(r - d/2)}{\Gamma(r)} (x - i0)^{d/2 - r}, \qquad \frac{T_r(x)}{T_1(x)} = \frac{\prod_{j=1}^{r-1} (d/2 - j)}{(r-1)! x^{r-1}}. \tag{74}$$

The second equality follows by applying $\Gamma(z+1) = z\Gamma(z)$ repeatedly to the first. The empty product at $r = 1$ is one. Applying Eq. (74) term by term in Eq. (69) proves both the reduction to unit tadpoles and their full analytic evaluation. The result is a meromorphic expression in d; ultraviolet or infrared poles are retained. The same tadpole value also evaluates any unit tadpole returned by the retained zero-invariant tensor formulas. A remaining master with several denominators still requires its own evaluation.

Independent components multiply, as proved in Appendix A. The direct implementation accepts one through thirteen such loop momenta, at most eight denominators per component, individual absolute powers at most eight, and total positive power after equal-mass combination at most 48 per component. Masses are nonnegative rational numbers. Denominators with nonzero target power must carry one of those individual momenta with coefficient $+1$ or -1. These tests exclude couplings between components and specify the actual scope of the rule.

The connected one-mass vacuum trees have a separate analytic evaluation. Appendix H derives their dimensionless Gamma product $C^V_{\mathcal{T}}$. With total physical power A, Wick rotation of all integer-power denominators gives

$$V_{\mathcal{T}} = (-1)^A (m^2 - i0)^{Ld/2-A} C^V_{\mathcal{T}}. \tag{75}$$

The exponent follows either from the final radial integral or from the mass dimension of the L measures and A denominators. For a unit-power master, $A = P = 2L - 1$. The application now returns its explicit Gamma arguments and scale exponent together with the rational reduction. This exposes the analytic master value already implied by the tree derivation and avoids a separate master evaluation for these families.

J Analytic inputs and checks

The two-mass vacuum grid contains one request at each $L = 1, \ldots, 13$. At loop order L, each independent component has squared masses $(h+2, h+7)$ with $h = 23011+L$. Its four final target power pairs are $(2,1), (3,1), (2,3), (3,-1)$, and each full target repeats the same pair across all L components. These are products of one-loop factors. The connected vacuum grid has 34 requests: chains at $L = 1, \ldots, 13$, stars at $L = 3, \ldots, 13$, and binary trees at $L = 4, \ldots, 13$. Their parent maps are $\pi(j) = j - 1$, $\pi(j) = 0$, and $\pi(j) = \lfloor (j-1)/2 \rfloor$, respectively. These starting orders omit coincident small-loop routings. The root squared mass remains symbolic. Each final request has root power two, root power three, root and final edge powers two, and root power four, with every other physical power one. The one-loop request instead uses powers two through five. The unit-power integral remains the master. The profiles are $23111 + L$, $23211 + L$, and $23311 + L$ for the three shapes and fix any auxiliary construction choices. These requests test preservation of the connected identity and the earlier symbolic single-loop dispatch.

The repeated-mass coverage grid uses four denominators per component with squared masses $(h+2, h+7, h+2, 0)$, where $h = 23511+L$. The four power assignments are $(1,1,2,0)$, $(2,1,1,1)$, $(1,2,0,-2)$, and $(0,0,0,2)$. This tests repeated masses, a massless pole, a polynomial numerator, and a scaleless target. It is checked through all thirteen loop orders, with no speed claim against an unfinished or unsupported native calculation.

The scaleless coverage grid has one family for each $L = 1, \ldots, 13$, with inverse propagators ℓ_i^2 and $(\ell_i - \ell_j)^2$ for all $i < j$. It therefore contains $L(L+1)/2$ coordinates and coupled loop momenta for $L > 1$. The three targets set all powers to one, raise the first power to two, or change the first power to minus one. Equation (67) supplies the reference zero. Native one- and two-loop checks test the same rule separately.

The three retained zero-invariant checks use the unequal, equal, and mixed mass choices of Eq. (22), with profiles $23211, 23212, 23213$, numerator rank two, and two excess positive powers. The second excess power is placed on the first physical denominator. The three ordinary tensor

checks use Eq. (23) with $h = 23311 + P$ for $P = 2, 3, 4$. Their numerator rank and excess positive power are both two, with one of the excess powers placed on the first physical denominator.

The new native reconstruction inputs are specified in Appendix K.

An independent symbolic partial-fraction calculation checks the vacuum grid. Random rational-function checks verify that subtracting the proposed poles leaves only a polynomial. Convergent Euclidean radial quadrature checks the tadpole normalization, and a native IBP comparison checks its raised-power recurrence. Native box and ordinary tensor coefficients are checked against independently prepared Kira reductions. The connected tree comparison checks preservation against the preceding exact rule; its mathematical justification is the separate derivation in Appendix H.

K Native reconstruction inputs

The planar and nonplanar boxes use Eqs. (24) and (25), respectively. The high-rank targets are $(1, 1, 1, 1, 1, 1, 1, -u, -v)$ with $u, v \geq 0$ and $1 \leq u + v \leq 4$. Each graph contributes fourteen training, six validation, and six reserved high-rank batches. The complete target sets differ between splits and exclude the preceding comparison batches. Individual target vectors can occur in more than one batch.

For batch index $i = 0, \ldots, 25$, define $j = i \bmod 6$ and $q = \lfloor i/6 \rfloor$. Let

$$\begin{aligned}(A_0, \ldots, A_5) &= (3, 13, 103, 1009, 10007, 100003),\\ (B_0, \ldots, B_5) &= (7, 19, 107, 1013, 10009, 100019).\end{aligned} \tag{76}$$

We set $s = -(A_j + 2q)$ and $t = -(B_j + 4q)$ in both graphs, using each graph's scalar-product convention. Indices 0–13 train the model, 14–19 select it, and 20–25 test it. These equations define the chosen input points. In the planar graph the fourth momentum is $-p_1 - p_2 - p_3$, and in the nonplanar graph it is $p_1 + p_2 - p_3$. Expanding their squares with the listed scalar products gives zero in both cases. The test batches are specified in Table 8.

Table 8: Reserved high-rank target batches. The same (u, v) sets and invariants enter both box routings. Every row defines a complete request for each graph.

i	s	t	Numerator powers (u, v)
20	-109	-119	$(2,0), (0,2), (0,1)$
21	-1015	-1025	$(2,2), (0,4)$
22	-10013	-10021	$(3,1), (3,0), (2,1), (1,3), (0,1)$
23	-100009	-100031	$(3,1), (2,2), (2,1), (2,0), (1,2), (0,4)$
24	-11	-23	$(2,2), (2,1), (1,2), (1,1), (0,2), (0,1)$
25	-21	-35	$(2,1), (1,3), (1,2), (1,0), (0,2), (0,1)$

For the lower-rank box requests, the sector corner is $(1, 1, 1, 1, 1, 1, 1, 0, 0)$. A numerator request uses the supplied first-rank auxiliary targets, and a raised-power request increases one physical index. Table 9 lists their precise target vectors. Their profiles are $h = 27701, 27713$ for training, 27733 for validation, and 27739 for testing, with the low-rank kinematics of Appendix D.

The sunrise family has loop momenta (ℓ_1, ℓ_2) and one external momentum p. Its inverse-propagator momenta and squared masses are

$$(Q_1, \ldots, Q_5) = (\ell_1, \ell_2, p - \ell_1 - \ell_2, p - \ell_1, p - \ell_2), \qquad (m_1^2, \ldots, m_5^2) = (3, 7, 13, 0, 0). \tag{77}$$

The first three coordinates are physical denominators and the last two are auxiliary. The three-loop vacuum uses

$$(Q_1,\dots,Q_6) = (\ell_1,\ell_2,\ell_3,\ell_1-\ell_2,\ell_2-\ell_3,\ell_1-\ell_3),$$
$$(m_1^2,\dots,m_6^2) = (1,1,1,13,13,2h+53). \tag{78}$$

All six coordinates are physical. Both structures use the same profile split as the low-rank boxes. For the sunrise, $p^2 = -1009-12h$. These assignments specify test families, rather than universal relations between masses and invariants. In every native request the kinematic quantities are fixed rational numbers and d remains symbolic, so the coefficients belong to $\mathbb{Q}(d)$. This test does not reconstruct general symbolic scattering invariants.

Table 9: Lower-rank reserved targets. Training and validation use the same index patterns at different physical points. A corner request contains the all-unit physical sector.

Structure and request	Target indices
Planar box, corner	$(1,1,1,1,1,1,1,0,0)$
Planar box, numerator	$(1,1,1,1,1,1,1,0,-1),(1,1,1,1,1,1,1,-1,0)$
Planar box, raised power	$(1,1,1,1,1,2,1,0,0),(1,1,1,1,1,1,2,0,0),(2,1,1,1,1,1,1,0,0)$
Nonplanar box, corner	$(1,1,1,1,1,1,1,0,0)$
Nonplanar box, numerator	$(1,1,1,1,1,1,1,0,-1),(1,1,1,1,1,1,1,-1,0)$
Nonplanar box, raised power	$(1,1,1,1,1,2,1,0,0),(1,1,1,1,1,1,2,0,0),(2,1,1,1,1,1,1,0,0)$
Sunrise, corner	$(1,1,1,0,0)$
Sunrise, raised power	$(1,2,1,0,0),(1,1,2,0,0),(2,1,1,0,0)$
Vacuum, corner	$(1,1,1,1,1,1)$
Vacuum, raised power	$(1,2,1,1,1,1),(1,1,2,1,1,1),(1,1,1,2,1,1)$

The protected native bases consist of the integral indices below. Each listed vector $\boldsymbol{b}$ denotes $M_{\boldsymbol{b}} = I_{\boldsymbol{b}}$ in its corresponding family. The independently prepared symbolic and reconstructed calculations use the same basis for each request. The lists specify the comparison basis; they do not assert a proof of global master independence from a finite seed system.

Planar box:

$$\{(0,0,1,1,0,0,1,0,0),(0,1,0,0,0,1,1,0,0),$$
$$(0,1,0,1,1,0,1,0,0),(0,1,1,1,0,1,1,0,0),$$
$$(1,0,1,1,1,0,0,0,0),(1,1,1,0,0,1,1,0,0),$$
$$(1,1,1,1,1,1,1,-1,0),(1,1,1,1,1,1,1,0,0)\}.$$

Nonplanar box:

$$\{(0,0,1,1,1,1,0,0,0),(0,1,0,0,1,1,0,0,0),$$
$$(0,1,1,1,1,1,1,0,0),(1,0,0,1,0,1,0,0,0),$$
$$(1,0,1,1,1,1,0,0,0),(1,1,0,0,0,0,1,0,0),$$
$$(1,1,0,1,0,1,1,0,0),(1,1,0,1,1,1,0,0,0),$$
$$(1,1,1,0,1,0,1,0,0),(1,1,1,1,0,0,1,0,0),$$
$$(1,1,1,1,1,1,1,-1,0),(1,1,1,1,1,1,1,0,0)\}.$$

Sunrise, basis 1:

$$\{(0,1,1,0,0),(1,0,1,0,0),$$
$$(1,1,0,0,0),(1,1,1,-1,0),$$
$$(1,1,1,0,-1),(1,1,1,0,0),$$
$$(1,1,2,0,0)\}.$$

Sunrise, basis 2:

$$\{(1,1,1,0,0)\}.$$

Three-loop vacuum, basis 1:

$$\{(0,1,1,1,0,1),(0,1,1,1,0,2),\\ (0,1,1,2,0,1),(1,0,0,1,1,1),\\ (1,0,1,1,0,1),(1,0,1,1,1,0),\\ (1,0,1,1,1,1),(1,0,1,1,2,0),\\ (1,1,0,1,0,1),(1,1,0,1,1,0),\\ (1,1,0,1,1,1),(1,1,1,0,0,1),\\ (1,1,1,1,0,0),(1,1,1,1,0,1),\\ (1,1,1,1,1,0),(1,1,1,1,1,1)\}.$$

Three-loop vacuum, basis 2:

$$\{(1,1,1,1,1,1)\}.$$

L Coefficient reconstruction

Kira and FireFly provide the finite-field algorithms used here [9, 10, 11]. To explain the relation to an exact rational coefficient, suppose

$$r(d) = P(d)/Q(d), \qquad P(d) = \sum_{j=0}^{n} p_j d^j, \quad Q(d) = \sum_{j=0}^{m} q_j d^j. \tag{79}$$

One nonzero denominator coefficient fixes the overall normalization. At a nonsingular point $d = z$ modulo a prime, a numerical reduction returns $r(z)$. Multiplying by the denominator gives the linear constraint

$$\sum_{j=0}^{n} p_j z^j - r(z) \sum_{j=0}^{m} q_j z^j = 0 \pmod{p}. \tag{80}$$

This follows directly from Eq. (79). With correct degree bounds and sufficiently many independent nonsingular samples, the constraints determine the normalized coefficients over the field. Combining suitable primes and reconstructing rational numbers then supplies rational coefficients [10, 11]. FireFly determines the necessary information adaptively; our model chooses the native procedure and probe batch size.

The field operations are exact, while reconstruction termination relies on its degree discovery and checks. Our evaluation therefore also subtracts every reconstructed coefficient from a separately computed symbolic reference and requires the rational difference to vanish. This checks the actual tested answer. Neither a model score nor a successful reduction to the supplied basis alone proves every coefficient for an arbitrary unseen request.

M Exact reduction of a one-scale massless triangle

The on-shell triangle has a Gamma-function representation [13]. We derive its coefficient in our normalization to specify the signs and the master basis used by the implementation. Let

$$T_{abc}(Q,d) = \int \frac{d^d k}{i\pi^{d/2}} \frac{1}{[k^2+i0]^a[(k-p_1)^2+i0]^b[(k-p_1-p_2)^2+i0]^c}, \tag{81}$$

where $p_1^2 = p_2^2 = 0$, $Q = -(p_1+p_2)^2 > 0$, and a, b, c are propagator powers. Write $A = a+b+c$ and $h = d/2$. For initially positive powers in a convergence region, Feynman parametrization and the shifted loop integral give

$$T_{abc} = \frac{(-1)^A \Gamma(A-h)}{\Gamma(a)\Gamma(b)\Gamma(c)} \int_{x_j \geq 0} d^3x\, \delta(1-x_1-x_2-x_3) x_1^{a-1} x_2^{b-1} x_3^{c-1} (Q x_1 x_3)^{h-A}. \tag{82}$$

The only nonzero squared momentum difference is between the first and third propagators, which gives the factor Qx_1x_3. The Dirichlet integral follows by writing three Gamma integrals and changing their variables to a common scale and three fractions. Explicitly, the substitution $t_j = ux_j$ in $\prod_j \int_0^\infty dt_j\, t_j^{z_j-1} e^{-t_j}$ gives

$$\int d^3x\, \delta(1-\textstyle\sum_j x_j) \prod_j x_j^{z_j-1} = \dfrac{\prod_j \Gamma(z_j)}{\Gamma(\sum_j z_j)}. \tag{83}$$

Applying this identity with $z_1 = h-b-c$, $z_2 = b$, and $z_3 = h-a-b$ yields

$$T_{abc} = (-1)^A Q^{h-A} \frac{\Gamma(A-h)\Gamma(h-b-c)\Gamma(h-a-b)}{\Gamma(a)\Gamma(c)\Gamma(d-A)}. \tag{84}$$

Dimensional continuation defines the expression outside the initial convergence region. The limit $b \to 0$ gives the two-point integral obtained by removing the middle propagator. Its unit-power value is

$$B(Q,d) = T_{101} = Q^{h-2} \frac{\Gamma(2-h)\Gamma(h-1)^2}{\Gamma(d-2)}. \tag{85}$$

For positive integers a, c and a nonnegative integer b, division of Eq. (84) by Eq. (85) gives a rational coefficient. To evaluate it with finite products, define

$$(z)_n = \frac{\Gamma(z+n)}{\Gamma(z)} = \begin{cases} \prod_{j=0}^{n-1}(z+j), & n \geq 0, \\ \dfrac{1}{\prod_{j=1}^{-n}(z-j)}, & n < 0. \end{cases} \tag{86}$$

The empty product equals one. The identity follows by repeated application of $\Gamma(z+1) = z\Gamma(z)$ and gives

$$T_{abc} = R_{abc}(Q,d) B(Q,d), \tag{87}$$

$$R_{abc} = \frac{(-1)^A Q^{2-A}}{(a-1)!(c-1)!} (2-h)_{A-2} (h-1)_{1-a-b} (h-1)_{1-b-c} (d-A)_{A-2}. \tag{88}$$

For example, $R_{111} = 2(d-3)/[Q(d-4)]$. Increasing a by one gives the recurrence

$$R_{a+1,b,c} = -\frac{(A-h)(d-A-1)}{Qa(h-a-b-1)} R_{abc}, \tag{89}$$

which supplies a separate exact check on the finite-product calculation. If $a = 0$ or $c = 0$, the remaining massless denominators have no nonzero external invariant. Those sectors vanish in dimensional regularization [1].

The implementation checks all masses and pairwise momentum differences before using this formula. It accepts rational $Q > 0$, powers from zero through four, and $A \leq 8$. It rejects cut propagators, additional occupied lines, and negative indices. For independent loop components with no occupied cross-loop denominator, the product measure then gives

$$\prod_{j=1}^{L} T_{a_j b_j c_j}(Q_j, d) = \left[\prod_{j=1}^{L} R_{a_j b_j c_j}(Q_j, d) \right] \prod_{j=1}^{L} B(Q_j, d). \tag{90}$$

The proof follows by factoring the integrand and measure in a common convergence region and then continuing the result analytically. If the requested integer powers have no common convergence region in d, we first allow the propagator powers to vary independently. Equation (84) gives their meromorphic continuation, which defines the product at the requested powers and dimension. This extension covers products through thirteen loops in the implementation. The implementation retains this product of component coefficients instead of expanding it into a larger polynomial. It supplies no formula for a general connected thirteen-loop scattering graph.

N Inputs for the final triangle and native comparison

The four native propagator structures are those in Appendix K. For the new planar and nonplanar box requests, the invariants are $s = -(137+6i)$ and $t = -(167+4i)$, with the incoming/outgoing momentum conventions stated there. Training uses $i = 2, \ldots, 13$, validation uses $i = 14, \ldots, 18$, and reserved tests use $i = 19, \ldots, 23$. Each box target has powers $(1, 1, 1, 1, 1, 1, 1, -a, -b)$. The reserved pairs are listed below; both box structures use the same pairs at each i.

i	Pairs (a, b) in the target request
19	$(3,0), (2,2), (0,2)$
20	$(4,0), (3,1)$
21	$(4,0), (1,2), (0,2), (0,1)$
22	$(3,1), (2,1), (0,2)$
23	$(2,1), (2,0), (1,0), (0,4), (0,2), (0,1)$

The sunrise masses remain $(3, 7, 13)$ and its external invariant is $p^2 = -386221 - 12i$. The three-loop vacuum masses remain $(1, 1, 1, 13, 13, 64255 + 2i)$. These families use the same index splits. Even i requests the sector corner, and odd i requests the raised-power targets defined for that structure. Full physical inputs differ between splits. Graph structures and individual index vectors recur; box target batches are distinct across the new splits.

For a triangle product with L loops, every factor has $Q = 101 + 2L$. We choose factor powers from $(1, 1, 1)$, $(2, 1, 1)$, $(1, 2, 1)$, $(1, 1, 2)$, $(2, 2, 1)$, $(1, 0, 1)$, and $(2, 0, 2)$. We form six products per loop order and remove duplicate full targets. Cross-loop auxiliary powers are zero. Seven additional single-loop requests use $Q = 211 + 2j$, $j = 1, \ldots, 7$, and the targets $(1, 1, 1)$, $(1 + (j \bmod 3), 1 + (j \bmod 2), 1)$, $(1, 0, 1)$, $(2, 0, 1)$, and $(0, 1, 1)$, again with duplicate targets removed.

The preceding triangle basis uses unit-power triangle masters. For verification, each such master is replaced by $2(d-3)B/[Q(d-4)]$, and independent components multiply. Kira provides a separate bubble-basis reference for each new single-loop request. The higher-loop comparison combines this exact basis conversion with the product derivation and independent Gamma-function checks.

References

[1] Stefan Weinzierl. *Feynman Integrals: A Comprehensive Treatment for Students and Researchers*. UNITEXT for Physics. Springer, 2022. arXiv:2201.03593; publisher DOI.

[2] K. G. Chetyrkin and F. V. Tkachov. Integration by parts: The algorithm to calculate β-functions in 4 loops. *Nuclear Physics B*, 192:159–204, 1981. publisher DOI.

[3] Stefano Laporta. High-precision calculation of multiloop Feynman integrals by difference equations. *International Journal of Modern Physics A*, 15:5087–5159, 2000. arXiv:hep-ph/0102033.

[4] Philipp Maierhöfer, Johann Usovitsch, and Peter Uwer. Kira—a Feynman integral reduction program. *Computer Physics Communications*, 230:99–112, 2018. arXiv:1705.05610.

[5] Fabian Lange, Johann Usovitsch, and Zihao Wu. Kira 3: Integral reduction with efficient seeding and optimized equation selection. *Computer Physics Communications*, 322:109999, 2026. arXiv:2505.20197.

[6] Zhuo-Yang Song, Tong-Zhi Yang, Qing-Hong Cao, Ming-xing Luo, and Hua Xing Zhu. Explainable AI-assisted optimization for Feynman integral reduction, 2025. arXiv:2502.09544.

[7] A. Denner and S. Dittmaier. Reduction schemes for one-loop tensor integrals. *Nuclear Physics B*, 734:62–115, 2006.

[8] Chang Hu, Bo Feng, Jiyuan Shen, and Yaobo Zhang. General one-loop generating function by integration-by-part relations. *Physical Review D*, 111:076026, 2025.

[9] Jonas Klappert, Fabian Lange, Philipp Maierhöfer, and Johann Usovitsch. Integral reduction with Kira 2.0 and finite field methods. *Computer Physics Communications*, 266:108024, 2021. arXiv:2008.06494.

[10] Jonas Klappert and Fabian Lange. Reconstructing rational functions with FireFly. *Computer Physics Communications*, 247:106951, 2020.

[11] Jonas Klappert, Sven Yannick Klein, and Fabian Lange. Interpolation of dense and sparse rational functions and other improvements in FireFly. *Computer Physics Communications*, 264:107968, 2021.

[12] NIST Digital Library of Mathematical Functions. Gamma function: Definitions and functional relations. Sections 5.2 and 5.5, definition and recurrence; accessed October 20, 2026.

[13] Andrei I. Davydychev and Per Osland. On-shell two-loop three-gluon vertex. *Physical Review D*, 59:014006, 1998.

[14] Leo Breiman, Jerome H. Friedman, Richard A. Olshen, and Charles J. Stone. *Classification and Regression Trees*. Chapman & Hall, 1984. Publisher record.

[15] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017. arXiv:1703.06114.

[16] Pierre Geurts, Damien Ernst, and Louis Wehenkel. Extremely randomized trees. *Machine Learning*, 63:3–42, 2006.

[17] Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. publisher DOI.

[18] Ben Ruijl, Takahiro Ueda, and Jos A. M. Vermaseren. Forcer, a FORM program for the parametric reduction of four-loop massless propagator diagrams, 2017.